

Aman Keshav Patre

Data Analyst — Data Scientist — Machine Learning Engineer
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Professional Summary

Computer Engineering graduate (CGPA: 9.16) with hands-on industry experience in **data science, predictive analytics, and process optimization** at UPL Limited. Skilled in Python, SQL, Power BI, and machine learning with a track record of building end-to-end ML pipelines, clustering models, analytics dashboards, and IoT-integrated data systems. Seeking entry-level roles in **data analytics or data science** where analytical rigor and production-ready data solutions drive business outcomes.

Experience

Data Science Intern

July 2025 – September 2025

UPL Limited, Gujarat — Energy Optimization & Video Analytics

- Cleaned and analyzed 3+ industrial IoT sensor datasets using Python (Pandas, NumPy), identifying 200+ anomalous records across shift-level production logs to surface actionable operational inefficiencies.
- Built optimization models across 6 operational parameters, improving plant energy efficiency KPIs and reducing manual reporting effort for weekly cross-departmental reviews.
- Developed 3 interactive Power BI dashboards tracking real-time energy consumption metrics, adopted by 2 engineering departments for daily operational decision-making.
- Annotated 1,000+ video frames with structured quality criteria to build training datasets for a computer vision analytics pipeline, maintaining defect-log documentation throughout.

Projects

Customer Segmentation & Predictive Analytics Engine

Python, K-Means, PCA, Streamlit

[Live Demo](#)

- Built an end-to-end unsupervised ML pipeline on 2,200+ customer records; applied K-Means clustering with Elbow Method hyperparameter tuning to identify $K = 6$ optimal segments, reducing misclassification across diverse behavioral profiles.
- Engineered 5+ behavioral features (Total_Spending, channel ratios, Age, platform Tenure) from raw transaction and demographic data; applied z-score normalization via StandardScaler to eliminate distance-metric bias.
- Deployed PCA to reduce a 7-dimensional feature space to 2 principal components, enabling visualization of 6 distinct customer personas and directly informing targeted go-to-market strategies.
- Shipped a production Streamlit dashboard with model caching (`@st.cache_resource`), churn-risk vs. premium-asset UI alerting, and `joblib`-serialized artifacts for zero-latency inference.

Smart Anna — Intelligent Cafeteria Management System

Python, IoT, Supabase (PostgreSQL), Vercel

[Live Demo](#) — Deployed on Vercel

- Developing an IoT-enabled full-stack platform with real-time occupancy dashboards across 5+ zone sensors, digital ordering system, and revenue trend analytics deployed on Vercel.
- Architected a Supabase (PostgreSQL) cloud backend supporting 200+ daily transactions, menu management, order tracking, and feedback collection with normalized relational schemas.
- Built analytics dashboards to visualize revenue trends, popular menu items, and operational KPIs for cafeteria management teams.

AgriYield Pro — Crop Yield Prediction Platform

Python, Streamlit, Random Forest, Scikit-learn

- Built an end-to-end ML web application predicting crop yield (hg/ha) across 10+ crop types using a production-ready Random Forest pipeline with cached model loading and label-encoded categorical features.
- Designed an interactive Streamlit dashboard with dynamic input forms, prediction history tracking, CSV export, and feature importance visualization to surface top yield drivers.
- Engineered the full ML lifecycle — data preprocessing, training, evaluation, and deployment — achieving consistent predictions across 5,000+ test samples.

EduPredict Pro — Student Performance Prediction System

Python, Scikit-learn, Streamlit, Plotly

[Live Demo](#)

- Trained a Linear Regression pipeline (One-Hot Encoding + StandardScaler) on 7 student features, achieving **88% prediction accuracy** for math score estimation across a multi-demographic dataset.
- Engineered 3 core application modules: instant single-prediction UI, batch CSV processing mode, and a Scenario Simulator to quantify the academic impact of test preparation interventions.
- Delivered interactive Plotly visualizations correlating reading and writing scores with math performance, and integrated direct email reporting for result distribution.

UIDAI Biometric Demand Intelligence Dashboard

Python, Streamlit, Predictive Analytics, Time-Series

[Live Demo](#) — Built for UIDAI National Hackathon

- Engineered an AI-powered workforce planning dashboard processing **1.8M+ simulation data points** to forecast Aadhaar biometric enrollment demands across multiple states with zero performance degradation.
- Implemented predictive demand forecasting with 12 unique regional forecasts, enabling data-driven allocation of biometric devices, operators, and facilities for national identity infrastructure.
- Built automated operational risk modeling to detect high-risk enrollment months, estimate localized infrastructure financial footprints, and prevent system bottlenecks at scale.
- Delivered an executive-grade interactive dashboard with granular demographic filtering (age cohorts: 5–17 and 18+), state-level geographic tracking, and real-time KPI indicators across 12+ regions.

Languages & Technologies

Languages	Python, SQL, MySQL
ML / AI	Scikit-learn, XGBoost, K-Means Clustering, PCA, Random Forest, Linear Regression, TensorFlow, PyTorch, Pandas, NumPy, SciPy, Seaborn, OpenCV, Matplotlib, Plotly
Analytics	Power BI, Streamlit, Excel (Advanced), EDA, Statistical Analysis, RFM Analysis, Cohort Analysis, A/B Testing
NLP	Text Preprocessing, TF-IDF Vectorization, Classification Pipelines, LLM Concepts
Engineering	Feature Engineering, Predictive Modeling, Unsupervised Learning, Data Pipelines, Optimization Modeling, Computer Vision
Cloud/Tools	Supabase (PostgreSQL), Vercel, Google Cloud, Git, GitHub, Jupyter Notebook, Google Colab, joblib, Dill, Streamlit
Coursework	Machine Learning, DBMS, Data Mining, Statistics, Artificial Intelligence

Education

B.E. Computer Engineering	May 2026
UPL University of Sustainable Technology, Gujarat	
CGPA: 9.16 — SPI (7th Semester): 9.25	

Certifications

- Developing Data Science Projects with Google Colab — Udemy (NLP, classification, end-to-end ML workflows)
- Data Analysis Using Python
- Deloitte Australia Data Analytics Job Simulation — Advanced analytics, business insight reporting